

Institute of Actuaries of India

Subject SA3 – General Insurance

November 2019 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i) There can be the following reasons behind this stand taken by the Reinsurer.

- The Reinsurer must have been suffering from huge claim outgo over a number of years coming out of the business written from the Industry and the prevailing premium rates are no longer sufficient to meet the claim pay-outs
- To gain market share, the direct insurers might be offering abnormal discounts to corporate clients which has led to a situation where the retail policyholders are subsidising the corporate clients. The reinsurer possibly wants to bring back discipline in the market place
- The regulatory scrutiny may have revealed this issue of under-pricing and the Reinsurer may have been asked to correct the price inadequacy
- The Reinsurer, if a listed entity, will try to improve its profitability in order to manage the shareholder's expectations

[0.5 mark for each point]

[Max 2]

ii) The exposure measure is usually the sum insured but it is complicated for two reasons in case of Commercial fire insurance:

- Since the amounts of stocks the insured is holding may vary considerably over the period of the insurance, stock may be covered on a declaration basis. This will usually be determined retrospectively and there will be an adjustment premium at the end of the year.
- There is no standard way of allowing for inflation in the policy. Hence policies with different types of inflation need to be considered separately to determine the exposure.

[1.5]

[1.5]

[3]

iii) The objective of the exercise is to estimate the Burning Cost (BC) as estimated future annual losses per unit of exposure. As this will be for Commercial fire insurance, the BC needs to be estimated at occupancy code level.

Assuming that the transaction level data is available for historical years, the analysis can be carried at policy year wise at the occupancy code level. Considering the context of the requirement, it would be better to analyse latest three years data to capture the recent trends in claims experience. As the data is available for the entire industry, there would not be any major issue with the volume. Key variables for the analysis are:

- Policy Year
- Policy Period
- Occupancy Code
- Sum Insured
- Nature of loss
- Claims paid and incurred (both including Allocated Loss Adjustment Expenses)
- Opening and closing outstanding for last three years
- Any policy level endorsements

[2]

Data preparation

The following steps need to be followed while preparing the data for the exercise.

There are likely to be three types of data sets – a) Policy data; b) Occupancy data & c) Claims data

The Policy data should contain the Policy Year, Policy Period, Premium etc.

The Occupancy data should contain the information related to SI & Occupancy Code. This needs to be merged with the policy data using some common variable like Policy Number.

[2]

On the other hand, the claims data should contain the claims information (paid and outstanding) along with some common variable to map the claims data to the policy level data and occupancy codes.

If ALAE is not included and recorded separately, the same needs to be added to the claim amount.

As any other actuarial exercise, it is imperative to reconcile the claims data – paid and incurred with any other industry level information available.

It is also possible that few policies will get excluded during the data preparation stage. Some examples can be (depending on the objective of the exercise):

- Policies those underwent cancellations
- Policies with period other than one year
- Coinsurance policies can also be removed if industry participants have shared data inconsistently and it is difficult to understand which is before applying coinsurance or after
- Policies with very high SI under any occupancy code (threshold can be defined and agreed at the outset)
- Any other data anomalies observed

[1]

All claims corresponding to the policies excluded (as mentioned above) are also to be excluded from the analysis.

Losses due to catastrophe perils are better to be excluded to arrive at the BC. Separate loading can be made for the expected catastrophe losses.

Once the data preparation is done, for the three policy years under consideration, total written exposure for different occupancy codes are to be calculated. Cumulative claims paid, incurred and the claim counts as at the end of the last three policy years then need to be found out for different occupancy codes. Losses can be capped at a pre-agreed threshold level.

SI and hence exposure during the policy period can be changed. Endorsement record would help capture these changes in SI. Exposure for the endorsement records can be taken as (Policy period left/total Policy period)* Change in SI for that record. In case the changes in premium due to the endorsements are higher than usual, a call needs to be taken whether to include such policies in the analysis or not.

Other adjustments can also be made where for any particular Occupancy code it is found that one insurer has provided cover for extremely high SI and experienced no claim, whereas this occupancy code does not have much exposure from other insurers. In such cases, the record can be excluded from the analysis.

- Methodology
- The analysis is on Policy Year basis
- Paid and incurred claim triangles are created at Occupancy code level and developed to ultimate values for losses capped at the threshold level
- As we are considering one-year policy period, exposures for first two years will be fully earned, while for the latest year, the earned exposure needs to be estimated
- Severity trend can be found out through data/market research
- Large loss loading can be applied using a valid large loss methodology
- ULAE not to be considered
- One rate for all SI band to be followed
- Suitable tail factor needs to be accounted for in the loss development analysis

Loss development

Cumulative paid and incurred claims development triangles need to be prepared for the three Policy Years. The triangles can be developed to the ultimate using Basic Chain Ladder method.

Using the suitably chosen development factors, ultimate losses need to be established for all Occupancies by Policy Year.

[1]

Earned exposure

Written exposure for annual policies are available. Since the latest Policy Year have not been fully earned, suitable assumption needs to be taken to estimate the earned exposure. Based on the spread of written premium across the Policy Year, such assumptions can be made. It can be assumed that the policies are written at the mid-point of the quarter under consideration.

[1]

Loss projection

Ultimate losses are to be projected for the future period where the BC will be applicable.

As this is related to Commercial fire insurance, Construction Industry Cost Index can be used for the purpose. Claims on-level factors for the three historical Policy Years are to be decided.

The next step is to calculate the ratio of claims to exposure for each Occupancy code separately. Sum of On-level ultimate claims from three Policy Years is then divided by the sum of earned exposure for the three Policy Years.

[1]

Credibility weighted BC

The methodology as outlined above provides the BC based on the observed experience. However, the volume of the experience for some of the Occupancy codes may not be sufficient enough to attach full statistical credibility. The objective is to arrive at credibility factors that could be attached to the BC based on the observed experience alone.

Any erstwhile tariff rate for the Occupancy codes can be considered as the Complement. But care must be taken to adjust for the impact of large losses, natural catastrophes, expenses, commission and profit margin included in the tariff rate. For example 40% of Tariff rate can be used as complement.

Suitable method from Empirical Bayesian Credibility Theory can be used like Bühlmann-Straub model to estimate the credibility factors.

[2]

Large loss loading

As the above analysis considers claims capped at a suitable threshold, a separate load will be required to account for the losses beyond this threshold. This can be done using Incurred claims as at the end of latest Policy Year. If a claim has incurred amount above the large loss threshold, the amount that is in-excess of threshold is considered. By Occupancy code, the ratio of Total incurred amounts in excess of the threshold to incurred amount (capped at the threshold) is considered. The selected large loss loading needs to be applied to the credibility-weighted BC as derived above.

[1]

Finally, the above estimated BC (which represents pure losses plus ALAE) should be loaded for Catastrophe losses, ULAE, expenses, commission, profit margin to arrive at the gross premium to be charged.

[1]

[Max 11]

iv) In case of a Commercial fire insurance policy, the actual premium charged is usually based on a combination of factors not limiting to the past experience, but also including client relationships and judgements by the Portfolio Manager and Case Underwriter. Some of the factors that drive this variation include

- Different profitability assessment
Two insurers with the same experience may have a very different view of the profitability of a portfolio depending on the methodology used – using P&L account, actual current year experience, assessment of true underlying experience etc. Prior year reserve releases and/or large claims and cat events can bring significant variation in any one year's experience compared to the true underlying experience.
- Different assumptions in Technical Pricing
There are a number of factors which can result in different but valid assessments of the technical premium like target return on capital, level of capital adequacy, level of expenses etc. Different assessments of the frequency and severity of large claims, natural catastrophes can also vary.
- Growth targets
Growth target based on written premium compared to targets based on profitability can result in unprofitable business being written.
- Insurance pricing cycle
Insurers behave differently when the prevailing rates are hard or soft. There may be an explicit business decision to move in and out of the market.
- Product strategy
The strategy adopted by a Portfolio Manager can have a significant impact on the premium rates. For example, there can be a business strategy to sell a product as a loss leader in the market.
- Existing relationships
As Commercial fire insurance is typically sold by intermediaries, the intermediaries can often influence the insurers to provide discounts on premium. This can vary between insurers.

[1 mark for each point]

[Max 4]

v) It is true that the Actuaries have been involved in the pricing of portfolios of small-to medium-sized risks for longer than they have been involved in pricing larger more complex risks.

Therefore, in insurance companies where actuaries give advice on the pricing of the smaller risks, their role is often well established and clearly understood by the management, underwriters and the actuaries themselves. There is often a degree of formality with pricing committees and procedure notes on how pricing reviews are to be carried out.

A further factor that drives the role and behaviour of the actuary in the pricing of portfolios of small- to medium-sized risks is that the technical rate is often

(assumed to be) known with a reasonable degree of accuracy. This is because the past experience of similar risks will be a reasonable guide to the future experience of the new risk.

[2]

In case of large commercial risks, there is always possibility of high degree of uncertainty regarding the technical rate. Pricing is heavily influenced by the underwriting cycle. Lack of structured data also put problem in actuarial pricing. Hence for such risks It is usually not possible to calculate a risk premium based on statistical analysis, instead a more reactive approach is taken to assess the adequacy of the rates charged on Commercial Lines portfolios.

[1]

[3]

vi) The following are some methods by which the rate adequacy in large commercial line portfolios can be carried out.

a. Rate change analysis

This is to directly monitor the change in rate over time. It is expected to be useful as information is available when risk is written rather than when claims start to emerge.

In case of auto-rated business, the old and new rating structure can be directly used on the portfolio. But in case of non auto-rated business, changes in business mix needs to be considered.

At the renewal stage, the rate change can be determined as $(\text{change in premium} / \text{change in exposure}) - 1$

b. Actual vs Technical rate analysis

The ratio of actual premium to technical premium needs to be calculated. This helps in monitoring any consistent overpricing or under-pricing in any specific risk.

c. Renewal vs Conversion rates

Renewal rate is calculated as number of renewals / no. of policies offered renewal. A high renewal may suggest that the insurer's premiums are low relative to their competitors.

On the other hand, the Conversion rate is calculated as number of policies converted / no. of quotes offered. Care needs to be taken where if the intermediaries know that the Insurer overprices in certain segments, they may not even request a quote, thereby distorting the analysis.

d. Loss ratio monitoring

Analysing the ultimate loss ratio (usually obtained through reserving analysis) it can be assessed whether the prevailing rates are adequate or not. This can also help in analysing emerging claims experience.

e. Lapse vs Renewal analysis

This compares historic loss ratio of lapsed and renewal business. This will give an idea of whether the renewed risks are a better risk than those that lapsed and also an idea of whether the existing rate increases are adequate for the new business-mix.

[4]

vii) The following are the various implications on the General Insurers in the country:

- It will bring some amount of discipline in the underwriting of the relevant risks under Commercial fire insurance as Insurers have to follow the suggested statistical principles to arrive at the premium rates for such industries.
- The effective premium increase would help the GI companies at a time when the Motor book is going through an extremely rough patch. This should help them to manage their financials to some extent depending upon the share of business in Commercial fire.

- Even if any Insurer does not want to place the reinsurance business with the said Reinsurer, other Reinsurers may also follow the same principle and the Insurers might find it difficult to obtain capacity at usual terms and conditions. This can happen if the large Reinsurer is the Market Leader in this case.
- This can also bring some future relationship issues between the Cedant and the Reinsurer.
- Long term broker and client relationship will also take a hit as the clients will find the premium rates too high as compared to the historical low rates they have enjoyed. Brokers may not be willing to place risks with the Insurers the way they used to do earlier.
- Regulator may seek justification from the Insurers who tries to deviate from this mechanism.
- In case other Reinsurers do not follow this, there may not be any capacity constraint in the market and the basic objective of this stand taken by one Reinsurer will not be met.
- The fallout of previous high level of discounts given to the Corporate clients has been borne by individuals, whose premiums have increased continuously. That has led to a huge distortion, with individuals and retail policyholders paying higher premiums per policy than corporate account holders with bargaining power. This issue is expected to reduce substantially.

[1 mark each for valid point]

**[Max 6]
[33 Marks]**

Solution 2:

i)

The following are the reasons for the variations:

- Different pricing methodologies: The underlying method of calculating prices for different companies might be different with different rating factors. [1]
- Policy conditions: Different companies might be offering different excesses, coverages or different types of cover [1]
- Different frequency and severity assumptions basis garage tie-ups underlying claims settlement philosophy [1]
- Pricing flexibility: Different companies might have different flexibility with regards to their prices [1]
- Preferred/Non-Preferred segments: The target markets for different companies are different and hence might offer differential rates basis what is preferred/non-preferred [1]
- For higher value cars the variation is much lower which could be because of more volatility of experience in lower value cars [1]
- Different companies would be at different stages of evolution and the relatively newer and aggressive players would be offering better rates to gain market access and share [0.5]
- Shareholder requirements/capital constraints of different companies are different which would reflect in the prices [0.5]
- Different reinsurance structures because of which risk appetite of insurance companies might be different [0.5]
- Differential commission structures- Some companies might have to lower companies if they are attracting higher volumes [0.5]
- Different expense loadings [0.5]
- Brand value: Some established companies might be charging higher premium basis the brand [1]
- Incorrect pricing/pricing logics [0.5]

[10]

ii)

The following will be used to justify the prices:

- Specify the assumptions used: frequency used, severity used, volumes assumed, expense loading, commission loadings [1]

- Since the company has some experience the same can be used as a starting point [0.5]
- Considering volumes are lower the experience might be volatile and hence the rating factors chosen and the assumptions chosen become critical [0.5]
- Would need to explain basis the data and assumptions why the other prices are not accurately reflecting the underlying risks [1]
- These are point estimates, the range of prices would also need to be specified for better understanding to differentiate basis various underlying/rating factors [1]
- Stress testing results would need to be explained and the impact of the same on the overall profitability and capital requirements would also have to be explained [0.5]
- Specify for what period the prices are valid and how frequently they are likely to change [0.5]

[5]

iii) Challenges in Product Structure:

- The maximum claim amount is INR 5000 while the average value of the toy is INR 7500. The insurance should ideally cover the full purchase value or replacement value. However, a maximum limit on the claim cost would help the insurer manage the claim payouts. [0.5]
- Upto 5 toys are insured. It is unlikely that all the 5 toys will be bought at one time. If they are bought at different times how will the insurance policy be modified to manage the same. From policy administration perspective this seems difficult. [1]
- The policy period is not defined. [0.5]
- If different toys are bought at different time points how will the overall coverage period coincide is not specified. [1]
- Maximum 2 claims seems a deterrent when 5 toys are covered [0.5]
- Once the toy is broken or damaged there might be a salvage value. How is the salvage value dealt with needs to be agreed upon. [0.5]
- Maximum Price of the product is fixed and as such there will be little leeway in changing the price [0.5]

[Max 3]

Challenges in Pricing:

- The price of INR 1,500 on an average toy of INR 7,500 seems very high. It is around 20% percent of the price of the toy. There will be a limited take up for the same considering the high price for the insurance [1]
- There is no experience available with the insurer as this a new product line. So assumptions would have to be made regarding the frequency of the claims to arrive at the price. The assumption regarding severity seems easier to make as the maximum claim amount is restricted to INR 5000 [1]
- The likelihood that a claim is made seems reasonably high as kids are prone to damage the toys. So the frequency of this cover is high. Assuming even a 25% frequency the risk cost of the same for a maximum amount of 5,000 is 1,250 which is ~83% of the maximum price charged [1.5]
- Once a claim is made and honored the probability that the second claim is made will be reasonably higher as this can change the customer behavior. As such the premium will be insufficient to meet just the claim costs [1.5]
- Considering that the product is new and no experience is available on the same, it will be difficult to find reinsurance on the same and the entire claims cost would have to be borne by the insurer [1]
- Considering that nearly the entire premium would be used to make claim payouts it will be difficult to cover expenses and commissions and can lead to losses for the insured [1]

- Flat premium structure would lead to skewness in the pricing. If low value toys are predominantly sold and insured then the cost of insurance would be a very high percentage of the price. Ideally the pricing should be as a percentage of the price of the toy.

[1.5]

[Max 7]

[10]

[25 Marks]

Solution 3:

- i) IFRS 17 standard will require best estimate of liability which means reserves need to be discounted. The introduction of discounting in setting reserves does not affect the underlying performance of the Insurer, it affects the reporting of results. Also it is to be noted that the proposed standard talks about specific risk adjustment for non financial risks which will be applied on top of the Best Estimate Liability to account for the risk of uncertain cashflows.

[2]

- ii) Two risk measures which can be used in determining the risk adjustment are

- a. Value at Risk (VaR)
- b. Tail VaR or Conditional Tail Expectation

[2]

Value at Risk provides a point measure of the downside potential for a given risk. On the other hand, the Tail VaR is the conditional mean of the loss distribution above a selected threshold.

VaR is not a coherent risk measure and does not obey the sub-additivity property. In the finance sector, VaR is commonly used as it is coherent when losses are normally distributed. However, in insurance, losses are generally positively skewed. A primary weakness of the VaR measure is that if the adverse outcome is worse than the assumed quantile cutoff, there is no way to measure how bad the outcome actually is.

Tail VaR, on the other hand, assigns equal weights to all outcomes beyond the prescribed threshold. It may be computationally challenging to calculate Tail VaR.

[2]

[4]

- iii) The key element is selecting the discount rate is the link between discount rate and the characteristics of the underlying cashflows. The following factors should be considered while deciding the suitable discount rate.

- **Prudence, given the assets held**

It is important that the insurer observes a degree of caution in deciding on the rate of discount. This should not exceed the return expected to be achievable on the investments. The insurer should take account of the types of investments and the fact that not all assets earn investment income.

- **Tax consideration**

Usually, the insurer does not need to deduct tax from the discount rate because claims are tax deductible.

- **Time consistency**

A fixed rate from year to year makes comparisons simpler, and would usually be the method favoured by accountants and auditors.

- **Type of business written**

General insurance business comprised of both short term liabilities and long term liabilities as well. In addition to this, the contract tenures are also varying. Long-tailed liabilities should be matched by long-term investments, suggesting that the return on those investments should be the basis of a suitable discount rate.

[1 mark each for valid points]

[Max 3]

iv)

- a) Level of aggregation refers to the grouping of insurance (or reinsurance) contracts into a single bucket. The portfolio should be an aggregation of contracts that are subject to similar risks, managed together and have been issued not more than a year apart.

[1]

As an example, in case of motor insurance, insurers can aggregate contracts into following portfolios –

- Only Third Party liability business
- Third Party, fire and theft as a second portfolio
- Comprehensive cover as a third portfolio

[1]

[2]

- b) Insurers use a various sophisticated approach to manage their risks, including sharing risks among policyholders and also minimizing the risk transfer from policyholders to the insurer. Following industry practices can be considered when assessing the level of aggregation.

Pooling of similar risks

Insurers usually take a large number of contracts of similar risks, and while doing that they set prices based on estimates of the expected severity of the risk. Losses on some contracts are compensated by profits on other contracts. The insurer thus spreads its risks among a large group of policyholders who are subject to similar risks.

Comparability

One of the key intentions behind IFRS 17 standard is to bring more transparency in the financial statements. But Industry participants will be also worried about the comparability between different insurers should there be any materially different way of aggregating contracts done by different insurers. As this is a principle-based approach, there is a high possibility of potential variation and insurers may want to come up with some form of consistent approach in level of aggregation.

[2]

The level of aggregation of insurance contracts will affect how the performance of the insurer will be reported in its financial statements.

The following issues can crop up with the level of aggregation requirements:

- Applying the annual cohorts' requirement would require significant changes to systems and increase costs;
- Currently profitability is monitored internally based on a higher level of aggregation than required by IFRS 17

This will also affect how contracts are identified compared to current practices, and may also affect the pricing of some contracts

[2]

[4]

- v) Sometimes regulatory guideline restricts the Insurer's ability to set a differential price or level of benefits for contracts or policyholders with different risk characteristics. For example, law or regulation might not allow gender discrimination even though the risks are known to be different. In grouping insurance contracts, IFRS 17 includes an exception to the overall grouping requirements that permits insurers to include such contracts in the same group.

[2]

[17 Marks]

Solution 4:

i)

	GWP	Reinsurance Premium	Expenses	Commission	Return	CAT XOL	XOL	Net Risk Premium
Motor	1,00,00,00,000	10,00,00,000	25,00,00,000	15,00,00,000	1,50,00,000	50,00,000	7,00,000	50,93,00,000
Commercial	50,00,00,000	42,50,00,000	12,50,00,000	11,00,00,000	6,37,50,000	25,00,000	3,50,000	-9,91,00,000
Total Business	1,50,00,00,000	52,50,00,000	37,50,00,000	26,00,00,000	7,87,50,000			
CAT RI		75,00,000						
XOL		10,50,000						
Total								41,02,00,000

Reinsurance Premium: 1.5 mark for Motor and Commercial + 1.5 mark for CAT RI And XOL

Expenses: 1 Mark

Commission: 1 Mark

Return Commission: 1 Mark

CAT XOL and XOL are calculated on a proportional business (In proportion of the GWP)

[6]

- ii) Sum Insured: 100 crores
Retention basis surplus lines: 10 crores
Quota Share: 40% retained – so retained risk: 4 crores
Portfolio XOL: 3.5 crores in excess of 0.5 crores
Ultimate Net retention = 50 lacs

[3]

- iii) Motor:
- Looks appropriate
 - The XOL would be used to manage the overall losses for luxury cars and total losses
 - This would reduce the volatility of the claims and overall capital requirements resulting from losses
 - The XOL is only for Motor OD. For TP liability is unlimited which is not reinsured. The liability arising from this book might erode capital
 - For OD book since premiums are generally sufficient to cover the losses high retention seems reasonable

[0.5 mark for each point]

[2]

Commercial:

- Looks inappropriate
- Large losses are protected
- The capacity seems reasonable to write smaller risks and hence overall large losses are managed
- However, the net risk premium is negative
- The return commission is lower than the direct commission
- For large losses the solvency needs to be demonstrated on gross losses allowing for 50% credit. Hence the benefit of additional 35% reinsurance would not be allowed for solvency purposes.

[0.5 mark for each point]

[2]

CAT XOL:

- Basis the capacity to write commercial risks it seems adequate
- It is used to manage large aggregate losses
- Level of reinstatements need to be considered

[0.5 mark for each point]

[1]

Portfolio XOL:

- Motor already has an XOL in place and hence should not be ideally covered
- Will reduce overall net losses further in line with the risk appetite of the company
- For a 100 crore risk of commercial the Ultimate net retention is effectively 50 lacs which defines the risk averse nature of the company
- The rate on line seems high

[0.5 mark for each point]

[1]

[6]

iv)

Factor based approach:

Merits:

- Simplicity: It is simple to understand and calculate
- Time tested: The approach has been used historically and has reasonably stood the test of time
- Accounting consistency: The approach is consistent with the local accounting practices
- It takes into account factors such as volume of business and claims along with the quantum of assets and reinsurance wherever applicable

[0.5 mark for each point]

[Max 2]

Demerits:

- The quantum of capital required is sensitive to parameters and the logic of the parameters is questionable. [1]
- Does not take into account all risks for e.g. credit risk is not considered [0.5]
- If premiums are reducing and loss ratios are fairly managed the solvency capital will reduce even if the exposures the company is taking may not reduce [1]
- For Tariffed products it will not reflect the underlying risks [0.5]
- For non-proportional reinsurance arrangements, the factor-based system would be inappropriate [0.5]
- A company having strong risk management practices would not get benefit under factor- based solvency regime [0.5]

Risk Based Capital Approach:

Merits:

- It is exhaustive and would consider all risks underlying the portfolio [1]
- It would encourage a culture of risk management [1]
- The structure of RBC is based on extensive studies and use cases and lends credibility to the same [0.5]

Demerits:

- Complex: The RBC calculations would be inherently complex in nature and time consuming [0.5]
- If internal models are used then they can be misused for capital calculations [0.5]
- Different approaches would lead to inconsistencies amongst companies [0.5]

[10]

[25 Marks]
